

Ch. 13 — P01 (A) Stochastic Gradient and Noise in SimOpt

Problem Bank | Chapter 13 | Type: A | Difficulty:

Chapter: 13 Type: A Difficulty:

Background

Simulation optimization (SimOpt) seeks to solve:

$$x^* = \arg \min_{x \in \mathcal{X}} g(x) = \mathbb{E}[G(x, \omega)]$$

where $G(x, \omega)$ is a single-replication output for scenario x with random seed ω . Because the expectation is approximated by a sample mean, gradient estimates are noisy.

Problem

A manufacturing system has a single tunable parameter: the reorder quantity $q \in [20, 200]$ units. After 30 replications at $q = 80$ and $q = 100$, you observe:

q	$\bar{G}(q)$ (cost/day)	$s(q)$ (std dev)
80	142.8	18.4
100	137.3	19.1

Tasks

(a) Estimate the finite-difference gradient:

$$\hat{\nabla} g \approx \frac{\bar{G}(100) - \bar{G}(80)}{100 - 80}$$

- (b) Construct a 95% confidence interval for the difference in means $g(100) - g(80)$ using a two-sample t -test with 29 degrees of freedom (each). At significance level $\alpha = 0.05$, is the gradient estimate statistically distinguishable from zero?
- (c) The finite-difference step $h = 20$ introduces **approximation bias** (because g is nonlinear) and **noise** (because samples are finite). Describe the trade-off: what happens to bias and noise as $h \rightarrow 0$? As $h \rightarrow \infty$?

- (d) Suppose you are allowed a total budget of $B = 300$ replications to evaluate the gradient at $q = 80$ and $q = 100$. If $\text{Var}[G(80)] \approx \text{Var}[G(100)] \approx \sigma^2$, show that the optimal budget allocation is equal ($n_1 = n_2 = 150$). What changes if $\sigma^2(80) \neq \sigma^2(100)$? Give the optimal allocation formula.

Reference values: $t_{0.025, 58} \approx 2.002$, $z_{0.025} = 1.960$.